ModelIt!

Student Activity Pack

How Does Your Phone Fit 10,000 Songs?

Modeling Digital Signal Processing, Data Compression, and Recommendation Algorithms

Grade 10L1 | NGSS HS-PS4-5, HS-ETS1-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think a song gets from a recording studio onto your phone?

2 What do you think happens to the sound quality when a music file is compressed to save space?

3 Draw a diagram showing how you think sound waves are converted into digital data.

4 How do you think Spotify or Apple Music knows what songs to recommend to you?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Audio Signal Quality | Data Compression Rate | Algorithm Accuracy | User Engagement | Recommendation Match

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High Quality

Audio Quality: Maximum | Compression: Minimal

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Compression

Audio Quality: Standard | Compression: Maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: New vs. Veteran User

Comparing algorithm with 0 vs. 10,000 hours of data

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Bit Depth — The number of bits used to represent each audio sample — 16-bit provides 65,536 possible values per sample while 24-bit provides over 16 million, capturing quieter details and greater dynamic range

Network Bandwidth — The speed of the data connection available for streaming, which limits the quality of audio that can be delivered in real time and determines whether compression must be increased

Listening Context — Environmental factors like background noise, earphone quality, and listener attention that affect whether compression artifacts are perceptible

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a device the size of your palm store an entire music library and somehow know what song you want to hear next? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the trade-off between file size and audio quality?**

2. **Why is it important that compression algorithms are based on human perception rather than pure mathematics?**

3. **What feedback loop drives recommendation algorithm improvement, and what are its limitations?**

4. **How would your model change if the target user was an audiophile versus a casual listener?**

5. **What ethical questions arise from algorithms that learn your preferences and influence your listening behavior?**

8 STEM Challenge: Design a Music Discovery Platform

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YOUR ENGINEERING MISSION

Design a music streaming system that optimizes the balance between audio quality, storage efficiency, and recommendation accuracy for a target user group.

THE SCENARIO

A startup is building a music app for students that must work on devices with limited storage (16GB) and unreliable cellular data. The app needs to store enough songs for offline listening while still delivering good sound quality and accurate recommendations. Your team must specify the compression settings, storage allocation, and recommendation algorithm approach.

Engineering Design Thinking

1. HOW MUCH STORAGE WILL YOU ALLOCATE FOR MUSIC VERSUS APP FUNCTIONALITY AND CACHED DATA?
2. WHAT AUDIO QUALITY SETTINGS WILL YOU OFFER AND WHAT ARE THE TRADE-OFFS OF EACH?
3. HOW WILL YOU HANDLE THE COLD-START PROBLEM WHEN A NEW USER HAS NO LISTENING DATA?
4. WHAT METRICS WILL YOU USE TO MEASURE WHETHER YOUR RECOMMENDATIONS ARE ACTUALLY GOOD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Audio Engineers design sound recording, processing, and playback systems using wave physics and digital signal processing. They work in music studios, tech companies (Spotify, Apple, Sony), and live sound, earning $50,000–$120,000/year. Data Scientists who build recommendation algorithms earn $90,000–$180,000/year at tech companies.